URMOSES

Urban Mobility Operating System for Enhanced Senior Transport

Master Business Plan

Los Angeles 2028 Olympic Games

Michelle Alter, CEO

Alex Luxxi, CTO

Folkert Meeuw, Strategic Advisor

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Executive Summary

URMOSES is the world's first solar-powered air-conditioned e-rickshaw fleet for barrier-free senior mobility. As a Social Enterprise, we solve the critical "last-mile gap" during the Los Angeles 2028 Olympic Games while simultaneously creating permanent legacy infrastructure for the city.

The Problem

During LA28, extensive "Green Zones" with car bans will be established. Seniors (Generation 62+) face massive mobility barriers between temporary public transit stops and stadium entrances. Conventional transportation solutions (taxis, Uber, buses) are unavailable in these zones or unsuitable for seniors.

Our Solution

URMOSES provides accompanied door-to-door transport through enclosed, air-conditioned e-rickshaws. The vehicles are powered by solar roof systems – a world first for zero-emission air conditioning without draining the drive battery.

Core Features:

- Solar-powered air conditioning (300-400W roof solar, 450W AC unit)
- Enclosed, weather-protected cabins
- Assisted boarding by trained "Mobility Buddies"
- Smart Points for barrier-free access (Tap-and-Go)
- No apps required – physical URMOSES Cards

Market Opportunity: LA28

The 2028 Olympic Games offer a unique showcase opportunity:

- 17 days of operation under extreme conditions (heat, crowds)
- Global media attention
- Demonstration of climate-neutral urban mobility
- Legacy infrastructure for Los Angeles post-2028

Financing

Component	Budget (USD)
100 Vehicles (Custom E-Rickshaw + Solar + AC)	1,200,000
50 Smart Points (Hardware + Installation)	1,500,000
Software/Cloud Infrastructure	1,000,000
Personnel (3 Years: 2025-2028)	3,000,000
Operations, Maintenance, Insurance	2,000,000
Total Budget	**8,700,000**

Financing Strategy (Hybrid Model):

- **Primary:** LA28 Committee (event financing), strategic sponsors (ESG-focused), City of Los Angeles (legacy infrastructure)
- **Secondary:** Impact investors (supplementary)

Team

- **Michelle Alter, CEO:** Community builder with 300,000+ followers, experienced in stakeholder relations and content creation
- **Alex Luxxi, CTO:** Technical systems architect with expertise in complex software and IoT systems
- **Folkert Meeuw, Strategic Advisor:** 30+ years experience in transport systems and fare collection, entrepreneurial background

Legal Structure

URMOSES will be structured as a Social Enterprise (B-Corporation / Benefit Corporation). This legal form combines mission-driven purpose with professional business management and enables access to impact investors while preserving the social mission.

Roadmap

- **Q2 2025:** Financing, partnerships (Cruise N Comfort USA)
- **Q3-Q4 2025:** Prototyping (solar-AC integration), Smart Point development

- **Q1-Q2 2026:** Pilot operations, team building, certifications
- **Q3-Q4 2026:** Fleet production (100 vehicles)
- **Q1-Q3 2027:** Infrastructure installation (50 Smart Points)
- **Q4 2027:** Test operations, personnel training
- **July 2028:** LA28 live operations (17 days)

The Problem: The Last Mile Gap

The Los Angeles 2028 Olympic Games present seniors with critical mobility barriers that cannot be addressed by conventional transportation solutions.

The Situation

Green Zones with Car Bans:

During LA28, extensive areas around venues will be closed to motorized individual traffic. Taxis, Uber, and private cars will have no access. This is a deliberate strategy by the LA28 Organizing Committee to reduce traffic chaos and emissions.

Temporary Public Transit Infrastructure:

Visitors will be directed to temporary Metro and bus stops. From there, they must cover the "last mile" to stadium entrances – typically 500 meters to 2 kilometers.

Extreme Climatic Conditions:

Los Angeles in July/August reaches temperatures of 30-35°C (86-95°F). For seniors, extended walks under direct sunlight pose health risks.

Why Conventional Solutions Fail

- **Walking:** Too long and too hot for seniors with limited mobility
- **Shuttle Buses:** Overcrowded, no individual assistance, boarding often too high
- **Wheelchairs/Electric Mobility Scooters:** No air conditioning, no weather protection, slow
- **Paratransit Services:** Limited capacity, long wait times, not scalable for events

The Consequence

Without adequate last-mile solutions, seniors remain excluded from Olympic experiences – despite having purchasing power and demonstrably higher interest in major cultural events than younger demographics. This contradicts the Olympic Movement's promise of inclusion.

The Solution: URMOSES

URMOSES (Urban Mobility Operating System for Enhanced Senior Transport) is an integrated system for barrier-free senior mobility, consisting of three core components:

1. Solar-Powered Air-Conditioned E-Rickshaws

Vehicle Platform:

Proven e-pedicab platform with custom modification:

- Enclosed cabin with doors and panoramic glazing
- Capacity: 2 passengers + companion
- Low entry with step
- Storage space for walking aids, rollators, personal items

The Technical Innovation: Solar Air Conditioning

URMOSES is the world's first e-rickshaw fleet with solar-powered air conditioning:

- **Solar Roof:** 1.5-2 m² high-efficiency modules (22% efficiency) produce 300-400W
- **Micro AC Unit:** 450W DC compressor (RIGID DV1910E-AC or Cruise N Comfort USA)
- **Separate Battery:** 12V LiFePO4 (1,000 Wh) stores solar energy
- **Energy Balance:** 180-270% coverage – surplus supports drive system

Why This Matters:

In Los Angeles, a 1.5 m² solar roof produces approximately 2,100-2,800 Wh per day during summer months (July/August) with 7+ peak sun hours. For 10 trips of 10 minutes AC operation, consumption is only 750 Wh. The massive surplus enables:

- Continuous operation without external charging of the AC battery
- Extended range for the drive system
- Zero-emission cooling (no grid electricity required)

2. Smart Points – Barrier-Free Access

URMOSES Smart Points are physical interaction points at public transit stops and stadium entrances:

- **Tap-and-Go Logic:** RFID/NFC-based identification without smartphone app
- **URMOSES Card:** Physical card (like transit ticket), available in advance
- **Real-Time Info:** Display of wait time until next available rickshaw
- **Privacy Protection:** Privacy filters prevent shoulder surfing in high-traffic zones
- **Weather Protection:** Canopy for waiting passengers

Technical Integration:

Smart Points are connected to the fleet via cloud backend and use open standards (OASC MIMs, NeTEx, SIRI) for seamless integration into city mobility apps.

3. Mobility Buddies – The Human Factor

Trained personnel accompanies every trip:

- Assistance with boarding and alighting
- Accompaniment to destination gate (not just curbside)
- Training in senior communication and emergency protocols
- Discretion for prominent guests (VIP service on request)

Market & Opportunity: LA28 as Showcase

Why Los Angeles 2028?

1. Ideal Testing Environment for Solar Technology

Los Angeles ranks among the sunniest metropolises worldwide:

- Average 6.6 peak sun hours per day (annual average)
- July/August: 7+ peak sun hours (Olympic period)
- Highest photovoltaic power output in North America (1,800 kWh/kWp)

2. Demographic Target Group On-Site

Los Angeles County has 1.8 million residents over age 62. This population group:

- Has above-average purchasing power
- Shows high interest in major cultural and sporting events
- Is currently excluded from many mobility offerings

3. Global Attention

The Olympic Games offer an unprecedented platform:

- 3+ billion global TV viewers
- 17 days of intensive media presence
- Showcase for sustainable urban mobility

4. City of Los Angeles as Strategic Partner

LA has committed to "Green Olympics":

- Zero-emission transport as priority
- Legacy infrastructure should remain usable beyond 2028
- Interest in solutions for aging population

Scalability Beyond LA28

While LA28 serves as showcase, URMOSES is designed from the start for replication:

- **Modular System:** Vehicles, Smart Points, and software are standardized
- **Licensing Model:** Other cities can adopt the URMOSES framework
- **Impact Proof:** LA28 delivers measurable data on usage, emission reduction, senior satisfaction

Business Model & Financing

Legal Structure: Social Enterprise

URMOSES will be founded as a B-Corporation (Benefit Corporation) in the USA. This legal form:

- Legally anchors social mission in articles of incorporation
- Simultaneously enables professional business management
- Qualifies for impact investors
- Protects against mission drift in later financing rounds

Hybrid Financing Model

The $8.7 million budget will be financed through three primary and one secondary source:

Primary Sources:

- **LA28 Organizing Committee (Event Financing):**
    - Integration as official Mobility Service
    - Marketing in Olympic packages
    - Estimated: $2-3 million
- **Strategic Sponsors (ESG-Focused):**
    - Mobility brands (e.g., e-bike manufacturers)
    - Insurance companies (health/senior care)
    - Payment providers (Visa – Tap-and-Go technology)
    - Estimated: $2-3 million
- **City of Los Angeles (Legacy Infrastructure):**
    - Co-financing Smart Points (remain after event)
    - Integration into city mobility budget
    - Estimated: $1-2 million

Secondary Source (Supplementary):

- **Impact Investors:** If primary sources do not fully cover, impact investors focused on Social Return on Investment will be approached (estimated: $1-2 million).

Revenue Streams (Post-Event)

After LA28, URMOSES can generate the following revenue streams:

- **Service Fees:** Usage-based billing per trip
- **Licensing:** Sale of URMOSES framework to other cities
- **Data Insights:** Anonymized mobility data for urban planning
- **Maintenance Contracts:** Long-term service agreements for infrastructure

Team & Organization

Management

Michelle Alter, CEO

Michelle is an established community builder with over 300,000 followers across various platforms. Her daily, unwavering presence and consistent content production demonstrate exceptional discipline and perseverance. As CEO, Michelle is responsible for:

- Stakeholder relations (LA28 Committee, City of LA, sponsors)
- Public relations & community engagement
- Strategic partnerships
- Corporate culture & mission guardianship

Alex Luxxi, CTO

Alex brings solid programming experience and the ability to quickly learn complex systems. Her thoughtful, analytical approach makes her the ideal architect for URMOSES' technical infrastructure. As CTO, Alex is responsible for:

- Development & maintenance of cloud infrastructure
- Integration of IoT components (Smart Points, vehicle telematics)
- Technical partnerships (Cruise N Comfort USA, solar manufacturers)
- Ensuring open standards (OASC MIMs, NeTEx, SIRI)

Folkert Meeuw, Strategic Advisor

Folkert brings over 30 years of expertise in transport systems and fare collection infrastructure. His entrepreneurial experience and deep understanding of railway and aircraft development enable well-founded strategic advice without operational interference. Folkert's role is deliberately limited to mentoring – Michelle and Alex make all business decisions independently.

Operational Structure

The following team will be built for LA28:

- **Site Management (1 Person):** Logistics coordination, interface to LA28

- **Fleet Management (3 People):** Maintenance, battery logistics, vehicle dispatch
- **Mobility Buddies (25-30 People):** Drivers & companions (core staff + flexible personnel for event peaks)

External Expertise

- **Legal Counsel:** Specialized in B-Corp formation and municipal transport law (USA)
- **Technology Partners:** Cruise N Comfort USA (air conditioning), solar manufacturers (PV modules)
- **IT Security:** External auditors for payment security (PCI-DSS)

Financial Planning

Investment Volume

The total budget for the LA28 project is $8.7 million and covers the period 2025-2028.

Component	**Budget (USD)**
Vehicles (100 Units)	
E-Pedicab Base ($6,000 × 100)	600,000
Custom Cabin + Glazing ($2,000 × 100)	200,000
Solar Modules ($1,500 × 100)	150,000
Micro AC Unit ($1,500 × 100)	150,000
AC Battery + Wiring ($1,000 × 100)	100,000
Subtotal Vehicles	**1,200,000**
50 Smart Points (Hardware + Installation + Licenses)	1,500,000
Software/Cloud Infrastructure	1,000,000
Personnel (Salaries 2025-2028, incl. Training)	3,000,000
Operations, Maintenance, Insurance	2,000,000
Total Budget	**8,700,000**

Note: This budget is conservatively calculated. An increase to $10-12 million would provide additional buffers for contingencies and an expanded fleet (120-150 vehicles).

Roadmap & Milestones

The project is structured in four phases:

Phase 1: Financing & Partnerships (Q2 2025)

- Completion of financing rounds (LA28, sponsors, City of LA)
- Contract with Cruise N Comfort USA (air conditioning)
- Solar manufacturer selection and partnership
- Formation of B-Corporation

Phase 2: Prototyping & Development (Q3-Q4 2025)

- Construction of 3 prototype vehicles (solar + AC integration)
- Development of Smart Point MVP
- Cloud backend & fleet management software
- Testing under LA summer conditions (energy balance validation)

Phase 3: Pilot Operations & Scaling (Q1 2026 - Q4 2027)

- **Q1-Q2 2026:** Pilot operations with 10 vehicles, team training, UL 2849 certification
- **Q3-Q4 2026:** Production of 100-vehicle fleet
- **Q1-Q3 2027:** Installation of 50 Smart Points, integration into LA Mobility Apps
- **Q4 2027:** Full-scale test operations, Mobility Buddies training

Phase 4: LA28 Live Operations (July 2028)

- 17 days of operations during Olympics
- Real-time data collection (usage, energy performance, satisfaction)
- Media showcase & global attention

Post-Event: Legacy & Scaling (from August 2028)

After LA28, Michelle & Alex will decide on the future direction:

- **Option A:** Continued operations in Los Angeles as permanent infrastructure

- **Option B:** Licensing of URMOSES framework to other cities
- **Option C:** Exit or pivot based on insights gained

Appendix: Technical Specifications

Vehicle Specifications

Component	Specification
Drive Power	500-750W electric motor
Speed	Max. 25 km/h (15 mph)
Payload	300 kg (2 passengers + luggage)
Range (Drive)	40-60 km per charge
Solar Air Conditioning	
Solar Roof	1.5-2 m², 22% efficiency, 300-400W peak
AC Unit	450W DC compressor (Cruise N Comfort / RIGID)
AC Battery	12V LiFePO4, 1,000 Wh (separate from drive)
Energy Balance (LA Summer)	2,100-2,800 Wh/day production, 750 Wh/day consumption

Technology Partners

- **Cruise N Comfort USA (Arizona):** Micro AC units for extreme conditions, Made in USA, proven with US Navy/Army
- **Solar Manufacturers:** High-efficiency modules (22%+), optimized for mobile applications

Compliance & Standards

- **UL 2849:** Electrical system certification (fire safety, mandatory from 2026)
- **PCI-DSS:** Payment Card Industry Data Security Standard (P2PE at Smart Points)
- **OASC MIMs:** Open & Agile Smart Cities Minimal Interoperability Mechanisms
- **NeTEx / SIRI:** European standards for timetable data and real-time information

- **GDPR / CCPA:** Data protection compliance (EU/California)

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URMOSES – Mobility as a Fundamental Right

Los Angeles 2028 & Beyond